MATH 110 — Week 5

Homework 5 — Solutions

HW5 P1(a).

Idea. The two evaluations force factors x and $x-1$; linear interpolation gives every output.

1. $Tp = 0$ iff $p(0) = p(1) = 0$. By the factor argument proved in Week 4, $p = x(x-1)q$ with $q \in \mathcal{P}_1(\mathbb{R})$. Conversely every such polynomial vanishes at both points.
2. Thus $\ker T = \operatorname{span}(x(x-1), x^2(x-1))$. The two polynomials are independent by their different degrees, so they form a basis.
3. For arbitrary $(a, b) \in \mathbb{R}^2$, the polynomial $p = a + (b-a)x$ has $Tp = (a, b)$. Hence T is onto.

HW5 P1(b,c).

Idea. The induced map records the two evaluations; cosets of $1, x$ already produce an output basis.

1. Let $K = \ker T$. The induced isomorphism is $\overline{T}(p + K) = (p(0), p(1))$. Its inverse sends (a, b) to $a + (b-a)x + K$, by part (a).
2. The cosets $1 + K, x + K$ map to $(1, 1), (0, 1)$, an independent spanning list of $\mathbb{R}^2$. An isomorphism preserves bases, so these two cosets are a quotient basis.
3. Equivalently, every $p + K$ equals $p(0)(1 + K) + (p(1) - p(0))(x + K)$, and evaluations force these coefficients uniquely.

HW5 P2.

Idea. The containment $U \subseteq \ker T$ ensures changing the representative does not change its image under T.

1. If $v + U = w + U$, then $v - w \in U \subseteq \ker T$, so $Tv = Tw$. Thus $\widetilde{T}(v + U) = Tv$ is well-defined.
2. $\widetilde{T}(a(v + U) + (w + U)) = T(av + w) = aTv + Tw$, proving linearity. Every quotient element is some $v + U$, so the prescribed formula also proves uniqueness.
3. A coset $v + U$ lies in the kernel iff $Tv = 0$, or $v \in \ker T$. Thus $\ker \widetilde{T} = \{v + U : v \in \ker T\} = (\ker T)/U$.

HW5 P3.

Idea. A single nonzero output can be scaled to every field element.

1. If φ is not zero, choose v with $a = \varphi(v) \neq 0$.
2. For each $b \in F$, $\varphi((b/a)v) = (b/a)\varphi(v) = b$. Hence φ is surjective.

HW5 P4.

Idea. Dual basis values isolate one coefficient at a time.

1. Write $v = \sum_j a_j v_j$. Applying φ_k gives $\varphi_k(v) = \sum_j a_j \delta_{kj} = a_k$, proving part (a).
2. For part (b), evaluate $\sum_j \psi(v_j)\varphi_j$ at v_k. The result is $\sum_j \psi(v_j)\delta_{jk} = \psi(v_k)$.
3. The two linear functionals agree on a basis, hence everywhere by Week 3 Proposition 21.

HW5 P5.

Idea. Entries of the original matrix are evaluations by the codomain dual basis.

1. Let $B = (v_1, \dots, v_n)$ and $C = (w_1, \dots, w_m)$, with dual bases φ_j and ψ_i. Since $Tv_j = \sum_i a_{ij} w_i$, we have $(T'\psi_i)(v_j) = \psi_i(Tv_j) = a_{ij}$.
2. The functional expansion gives $T'\psi_i = \sum_j a_{ij}\varphi_j$, so entry (j, i) of its matrix is a_{ij}.
3. This is exactly the transpose matrix A^{T}.

HW5 P6.

Idea. Write the vanishing conditions on the two generators.

1. A row (a, b, c) lies in U^0 iff $a + b = 0$ and $b + c = 0$, so it equals $b(-1, 1, -1)$. Thus $U^0 = \operatorname{span}((-1, 1, -1))$ over $\mathbb{R}$.
2. The generators of U are independent because the first coordinate of a zero relation forces the first coefficient zero, and the third coordinate forces the second zero. Hence $\dim U = 2$.
3. The annihilator generator is nonzero, so $\dim U^0 = 1$. Therefore $\dim U + \dim U^0 = 2 + 1 = 3$.

HW5 P7 (optional).

Idea. Separate a nonzero vector by one functional, then use finite-dimensional equality of dimensions.

1. For fixed v, evaluation $j_V(v)$ is linear in the functional; the identity $j_V(av + w)(\varphi) = a\varphi(v) + \varphi(w)$ also makes j_V linear.
2. If $v \neq 0$, extend v to a basis and define φ to be 1 on v and zero on every other basis vector. Finite basis expansions define a linear functional; arbitrary-dimensional basis extension uses the cited choice result.
3. Then $j_V(v)(\varphi) = 1$, so $j_V(v) \neq 0$, proving zero kernel and injectivity.
4. If V is finite-dimensional, $\dim V'' = \dim V' = \dim V$. The equal-dimensional injectivity criterion gives an isomorphism.

Exam practice, Question 2(a,b).

Idea. The dual map precomposes a functional with the original map.

1. Define $T^*(\varphi) = \varphi \circ T$ for $\varphi \in W^*$. It lies in V^* since composition of linear maps is linear.
2. For $a \in \mathbb{R}$ and $\varphi, \psi \in W^*$, $T^*(a\varphi + \psi)(v) = a\varphi(Tv) + \psi(Tv) = (aT^*\varphi + T^*\psi)(v)$ for all v. Thus T^* is linear.
3. For $\eta \in Z^*$, $(S \circ T)^*\eta = \eta \circ S \circ T = T^*(S^*\eta)$ by associativity, proving the product identity.

Exam practice, Question 2(c).

Idea. Surjectivity makes vanishing on all Tv the same as vanishing on all of W.

1. If $T^*\varphi = 0$, then $\varphi(Tv) = 0$ for all $v \in V$.
2. For each $w \in W$, choose v with $Tv = w$. Then $\varphi(w) = 0$, so φ is the zero functional.
3. Thus $\ker T^* = \{0\}$, and the zero-kernel criterion proves injectivity.

Exam practice, Question 2(d).

Idea. Compose the given scalar functional with the map.

1. $(T^*\ell)(x, y) = \ell(T(x, y)) = 3(2x - y) = 6x - 3y$.
2. Thus $T^*\ell$ is the functional represented by row $(6, -3)$.